

HARGAISA TEXT

Nidaamka Maamulka Cashuurta iyo Hantida Dawladda Hoose ee Hargeysa

1. Sharaxaadda Project-ka

Dulmar kooban: Hargaisa Text waa nidaam web ah oo shaqaalaha la oggolaaday ka caawiya maamulka guryaha iyo dhulka, xogta macaamiisha, goobaha khariidadda, warbixinnada iyo hawlaha maamulka. Qaybta hore waa React/TypeScript, adeegga dhexe waa Express, xogtana MySQL ama MariaDB ayaa kaydisa.

Dhibaataada uu xallinayo: Wuxuu hal meel ku ururiyaa xog markii hore ku kala jiri lahayd faylal ama habab kala duwan. Sidaas waxaa yaraada xog-lumid, isku dhex-yaac iyo raad-raac la'aan.

Ujeeddada: In la helo hab ammaan ah oo lagu diiwaangeliyo, lagu raadiyo, lagu cusboonaysiiyo, lagu khariidadeeyo laguna soo saaro warbixinta hantida iyo macaamiisha.

1.1 Dadka Isticmaali Kara

- Maamule (Admin): wuxuu heli karaa dhammaan qaybaha, isticmaalayaasha, diiwaanka hawlaha iyo kayd-celinta.
- Wakiil (Agent): wuxuu ku dari karaa ama wax ka beddeli karaa hantida iyo macaamiisha, balse ma tirtiri karo mana geli karo qaybaha maamulka gaarka ah.
- Daawade (Viewer): wuxuu akhrin karaa xogta la wadaago, laakiin wax kama beddeli karo.

1.2 Qaybaha Ugu Muhiimsan

- Gelitaanka iyo kala xaddididda awoodda isticmaalayaasha; dashboard muujinaya tirooyin iyo isbeddello.
- Maamulka hantida, sawirrada iyo macaamiisha; raadinta, kala shaandhaynta iyo bog-bog u bandhigidda xogta.
- Khariidadda GIS/Leaflet, warbixinno CSV, Excel iyo PDF, iyo ogeysiisyo.
- Profile, dejimaha qofka, maamulka users-ka, activity log, iyo backup/restore.
- Tax Management-ku wuxuu sameeyaa biil tusaale ah oo ah 1% qiimaha hantida; ma aha qiimayn cashuureed rasmi ah.

1.3 Sida Nidaamku u Shaqeeyo

1. Isticmaaluhu wuxuu geliyaa username iyo password; nidaamku wuxuu xaqiijiyaa akoonka.
2. Markuu galo, wuxuu arkaa qaybaha uu role-kiisu u oggol yahay.
3. Wuxuu raadiyaa ama diiwaangeliyaa hanti/macaamiil, ku daraa sawir ama goob khariidadeed, dabadeed xogta ayaa la hubiyaa oo la kaydiyaa.
4. Natiijada waxaa lagu arkaa dashboard, jadwal ama khariidad; warbixinna waa la soo dejisan karaa.

1.4 Faa'iidooyinka

Nidaamku wuxuu keenaa xog mideysan, shaqo degdeg ah, kala xaddidid masuuliyadeed, raad-raac hawleed, warbixin fudud, iyo backup lagu soo celin karo database-ka.

2. Sharaxaadda SRS-ka

Waa maxay SRS? SRS (Software Requirements Specification) waa dukumenti si cad u sheegaya waxa software-ku qabanayo, cidda isticmaalaysa, xogta uu kaydinayo, ilaalintiisa iyo xaddidaadihiisa. Af fudud, waa heshiis sharaxaya sida nidaamka loo fahmayo loona qiimaynayo.

Sababta loo diyaariyay: SRS-kan wuxuu macallinka, supervisor-ka, client-ka, horumariyaha iyo tester-ka siinayaa tixraac isku mid ah. Waxa ku qoran waxaa laga xaqiijiyey source code-ka, database schema-da, interface-ka, tests-ka iyo README-ga.

2.1 Waxyaabaha Muhiimka ah ee SRS-ka

- Ujeeddo, baaxadda project-ka, qaybaha nidaamka, dadka isticmaala iyo habka uu u shaqeeyo.
- Functional requirements: gelitaan, dashboard, property/client CRUD, sawirro, GIS, reports, notifications, settings, users, audit, tax PDF iyo backup.
- Non-functional requirements: ammaan, waxqabad, fudayd isticmaal, isku-hallayn, dayactir fudud, la jaanqaadid, ballaarin iyo soo-celinta xogta.
- Use cases, user-flow, database ERD, interface-yada, tijaabooyinka, xaddidaadaha iyo horumarinta mustaqbalka.

2.2 Roles, Database iyo Security

Roles-ka waa Admin, Agent iyo Viewer; API-gu ayaa mar kasta xaqiijiya awooddooda. Database-ku wuxuu leeyahay jadwallo ay ka mid yihiin users, clients, properties, property_images, notifications, user_settings, activity_logs iyo rate_limit_buckets. Xiriirrada iyo validation-ku waxay yareeyaan xog khaldan.

Ilaalinta xogta waxaa ka mid ah password la qariyey (bcrypt), JWT cookie, CSRF protection, CORS, Helmet, rate limiting, hubinta input-ka iyo activity log. Production-ka waa inuu adeegsadaa HTTPS iyo secrets ammaan ah.

2.3 Xaddidaadaha iyo Waxyaabaha Mustaqbalka

Hadda ma jiro hab dhamaystiran oo qiimayn rasmi ah, lacag-bixin, rasiid, ganaax, dayn cashuureed ama racfaan. Forgot-password delivery, SLA, data-retention policy, WCAG certification iyo backup jadwalaysan sidoo kale lama qeexin. Mustaqbalka waxaa lagu dari karaa qaybahaas, citizen portal iyo off-site backup.

Qodobbada Muhiimka ah ee Lagu Sharxi Karo Project-ka

- Project-ku waa nidaam full-stack ah: browser, API iyo database ayaa wada shaqeeya.
- Saddex role ayaa xaqiijiya in qof kastaa qabto hawsha loo oggolaaday.
- Hantida, macaamiisha, sawirrada iyo goobaha khariidadda hal meel ayaa laga maamulaa.
- Warbixinnada CSV, Excel iyo PDF waxay fududeeyaan gudbinta iyo falanqaynta xogta.
- Security, activity log iyo backup waxay ilaaliyaan xogta oo fududeeyaan raad-raaca.
- Biilka 1% waa tusaale; nidaam cashuureed sharci ah oo buuxa waa horumarin mustaqbal.

2.4 Gunaanad

Hargaisa Text waa project si cad u muujinaya maamulka hantida, kala xaddididda users-ka, khariidaynta, warbixinta iyo ilaalinta xogta. SRS-ku wuxuu caddaynayaa waxa hadda shaqaynaya iyo waxa aan weli la hirgelin, sidaas darteed qofka qiimaynaya wuxuu helayaa sawir buuxa oo sax ah.